Performance and Robustness of Signal-Dependent vs. Signal-Independent Binaural Signal Matching with Wearable Microphone Arrays

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***Abstract*—The increasing popularity of spatial audio in applications such as teleconferencing, entertainment, and virtual reality has led to the recent developments of binaural reproduction methods. However, only a few of these methods are well-suited for wearable and mobile arrays, which typically consist of a small number of microphones. One such method is binaural signal matching (BSM), which has been shown to produce high-quality binaural signals for wearable arrays. However, BSM may be suboptimal in cases of high direct-to-reverberant ratio (DRR) as it is based on the diffuse sound field assumption. To overcome this limitation, previous studies incorporated sound-field models other than diffuse. However, performance may be sensitive to signal estimation errors. This paper aims to provide a systematic and comprehensive analysis of signal-dependent vs. signal-independent BSM, so that the benefits and limitations of the methods become clearer. Two signal-dependent BSM-based methods designed for high DRR scenarios that incorporate a sound field model composed of direct and reverberant components are investigated mathematically, using simulations, and finally validated by a listening test, and compared to the signal-independent BSM. The results show that signal-dependent BSM can significantly improve performance, in particular for high DRR and with a rotation of the listener's head, while showing robustness to errors in the estimation of the source direction. Under low DRR conditions, performance of the signal-dependent methods become comparable to BSM, while still providing improvement with listener's head rotation.**

***Index Terms*—Binaural Reproduction, Wearable Arrays, Binaural Signal Matching, spatial encoding.**

I. Introduction

The binaural reproduction of acoustic scenes captured by microphone arrays is gaining popularity in diverse applications, such as teleconferencing and virtual and augmented reality, as evident by recent publications [1]–[3]. A common technique for binaural reproduction involves convolving high-order Ambisonics (HOA) signals with the head-related transfer function (HRTF) [4]. This method is relatively precise, especially for sufficiently high spherical harmonics (SH) orders, and it can provide a realistic immersion experience through the integration of head tracking [2]. However, the primary constraint of this method is the requirement for an extensive number of microphones when the audio signal is captured using a microphone array, as well as the need for a spherical array geometry, which limits the practical application of this method [2].

To address the limitations of binaural signal reproduction with spherical array geometries, the beamforming-based binaural reproduction (BFBR) technique was proposed [5]–[7]. In this method, microphone signals are filtered using a set of beamformers, followed by filtering with the head-related transfer functions (HRTFs), finally added to produce the binaural signals. This method is also commonly known as "virtual loudspeakers decoding" [8], [9]. A theoretical framework for determining the design parameters of BFBR, such as the number of beams and their look direction for both planar and spherical arrays, was presented in [10]. However, the guidelines for more general array geometries were limited and did not guarantee accurate binaural signal reproduction, leaving a comprehensive design methodology unavailable.

In an effort to overcome the limitations of existing beamforming-based methods and to accurately reproduce binaural signals captured by arrays of arbitrary geometry, a number of techniques have been developed [11]–[15]. The binaural signal-matching (BSM) technique is one such technique [16], [17]. The core principle of BSM is the design of a set of filters operating on the

microphone signals, with output that approximates the binaural signal. At the design stage, this is achieved by assuming a signal originating from a diffuse sound field, and minimizing the MSE between the filtered array signals and the desired binaural signal. The solution is then formulated in terms of the transfer functions of the array, i.e. the steering function, and the HRTF. It can therefore be considered as a signal-independent design, although it is based on the diffuse field assumption.

To enhance perception, at high frequencies, Magnitude Least-Squares (MagLS) optimization is used instead [17]–[19]. A BSM design was recently presented for a semi-circular array with varying numbers of microphones, incorporating head tracking [16], [17]. The study revealed that BSM accuracy heavily depends on microphone positioning, leading to poor performance when a listener's ear is distant from the recording microphones. In addition, the BSM which is designed under a diffuse sound-field assumption may be sub-optimal under high Direct-to-Reverberant Ratio (DRR) conditions, for example [20].

Parametric spatial audio was also investigated as an alternative for binaural reproduction. In these approaches, the sound field is decomposed into direct and reverberant components, each estimated and reproduced independently [21]–[25]. While this approach has shown promising results, its accuracy depends on the estimation of signal-dependent model parameters, such as the direction-of-arrival (DOA) of sound sources, and the direct-to-reverberant ratio (DRR) of the sound field. This uncertainty in the performance of the parametric approach defines a current gap in BSM-based binaural reproduction, where the choice of what is preferable - the robustness of signal-independent BSM or the performance of signal-dependent BSM, may not be entirely clear. Recent studies have shown benefit to the parametric methods over signal-independent BSM methods. Stahl [26] evaluated the performance of parametric approach implementation over scene-agnostic binaural reproduction methods, with static and rotated arrays, and for multiple array configurations. McCormack [27] further studied the performance under listener movement (6DoF). While these studies presented comprehensive performance analysis, their generalization to a wide range of conditions is yet to be explored.

This paper aims to address this gap by systematically investigating the incorporation of signal information into BSM. To diversify the study, two signal-dependent methods are investigated. One incorporates signal information in the correlation matrix employed in the BSM design as reported in [20], and the other is based on the parametric spatial coding of Coding and Multidirectional Parametrization of Ambisonic Sound Scenes (COMPASS) [25], which decomposes the signal into dominant sources and a residual reverberant component, and which was found to have superior performance [26].

Performance of these methods is investigated and compared to the signal-independent BSM. While single-source parametric methods are well-established (e.g. DirAC [21]), this study employs a single source for simplicity, as it has more focused aims - to study whether the additional computational complexity of parametric methods over the signal-independent BSM is justified, and under what conditions. We therefore specifically examine varying DRR and DOA estimation errors to identify the conditions where the advantage of the parametric methods may diminish, and whether under these conditions performance converges to that of BSM. Consequently, we restrict the study to a single speaker to isolate these trade-offs, acknowledging that this limits the study's generalization to challenging multi-speaker environments. The simulation study is based on an array mounted on a rigid sphere with a configuration that resembles the augmented-reality (AR) glasses, part of the EasyCom database [29], and is complemented by a listening test. Important conclusions arise from this study that can be useful when incorporating these methods:

1) A clear improvement in performance is observed for the signal-dependent BSM methods, in particular under head rotations, where one ear is typically rotated away from the array, and under high direct-to-reverberant ratio (DRR) conditions.
2) When source direction estimation is poor, the signal-dependent approaches degrade in performance but only to match that of the signal-independent BSM.

These results reinforce the confidence in the signal-dependent BSM approaches, showing relative robustness by not degrading below the signal-independent BSM under conditions of error in parameter estimation.

II. Mathematical Background

A. Signal Model

This subsection provides the mathematical model of the signal as used in this paper. Throughout the paper, the standard spherical coordinate system is used, denoted by (r, θ, ϕ), where r is the distance to the origin, θ is the inclination angle measured from the Cartesian z

axis downwards to the Cartesian xy plane, and ϕ is the azimuth angle measured from the positive x axis towards the positive y axis. We denote $k = \frac{2\pi}{\lambda}$ as the wave-number, where λ is the wave-length, f is the frequency and $\Omega = (\theta, \phi)$ represents the arrival direction of a plane wave in the sound field. Further, assume that the sound-field is composed of L far-field sound sources generating plane waves with arrival directions $\Omega_l, l = 1, ..., L$, and source signals $s_l(f), l = 1, ..., L$ [30]. The sound-field is captured by an array of M microphones, which are located at $(r_m, \Omega_m), m = 1, ..., M$, centered at the origin. The noisy array measurements can therefore be described by the following narrow-band model [30]:

$$\mathbf{x}(f) = \mathbf{V}(f)\mathbf{s}(f) + \mathbf{n}(f) \tag{1}$$

where $\mathbf{x}(f) = [x_1(f), x_2(f), ..., x_M(f)]^T$ is the microphone-signal vector (measurements), $\mathbf{V}(f) = [\mathbf{v}(k, \Omega_1), \mathbf{v}(k, \Omega_2), ..., \mathbf{v}(k, \Omega_L)]$ is an $M \times L$ complex matrix with columns $\mathbf{v}(k, \Omega_l)$ representing the array steering vector from the l-th source to the microphone positions for all $l = 1, ..., L$ sources, $\mathbf{s}(f) = [s_1(f), s_2(f), ..., s_L(f)]^T$ is the source-signal vector, and $\mathbf{n}(f)$ is an additive-noise vector.

In the reproduction stage we further assume that a listener is virtually positioned with the center of the head coinciding with the origin, where $h^{l,r}(k, \Omega)$ denotes the HRTF of the left and right ears of the listener using the superscripts $(\cdot)^l$ for the left ear and $(\cdot)^r$ for the right ear. The signal at the left and right ears can now be written as [31]:

$$p^{l,r}(f) = [\mathbf{h}^{l,r}(f)]^T \mathbf{s}(f) \tag{2}$$

where $\mathbf{h}^{l,r} = [h^{l,r}(\Omega_1), h^{l,r}(\Omega_2), ..., h^{l,r}(\Omega_L)]^T$ contains the HRTFs corresponding to the directions of the sources.

B. Binaural Signal Matching

This subsection provides the mathematical model of BSM. First, assume that the configuration of the microphone array is known, such that the steering matrix $\mathbf{V}(f)$ can be generated by simulation or measurement. In the first step, the array measurements are filtered and combined, in a similar manner to beamforming:

$$\hat{p}^{l,r}(f) = [\mathbf{c}^{l,r}(f)]^H \mathbf{x}(f) \tag{3}$$

where $\mathbf{c}$ is an $M \times 1$ complex vector holding the filter coefficients. Next, $\mathbf{c}$ is chosen to minimize the following mean-squared error between $\hat{p}^{l,r}(f)$ and $p^{l,r}(f)$, the binaural signals in (2), for each ear separately:

$$err^{l,r}_{bin}(f) = \mathbb{E}[|p^{l,r}(f) - \hat{p}^{l,r}(f)|^2] \tag{4}$$

where $\mathbb{E}[\cdot]$ is the expectation operator. Next, assume that the noise is uncorrelated to the sources. This leads to the following error formulation [16]:

$$\begin{aligned} err^{l,r}_{bin}(f) = \mathbb{E}[&\|[\mathbf{s}(f)]^H([\mathbf{V}(f)]^H \mathbf{c}^{l,r}(f) - [\mathbf{h}^{l,r}(f)]^*)\|_2^2 \\ &+ \|[\mathbf{n}(f)]^H \mathbf{c}^{l,r}(f)\|_2^2] \end{aligned} \tag{5}$$

Eq. (5) can also be written in a matrix form, omitting the explicit dependence on frequency, as:

$$\begin{aligned} err^{l,r} = &((\mathbf{c}^{l,r})^H \mathbf{V} - [\mathbf{h}^{l,r}]^T)\mathbf{R_s}(\mathbf{V}^H \mathbf{c}^{l,r} - [\mathbf{h}^{l,r}]^*) \\ &+ (\mathbf{c}^{l,r})^H \mathbf{R_n} \mathbf{c}^{l,r} \end{aligned} \tag{6}$$

Minimizing the error in Eq. (6) leads to the following solution [17]:

$$\mathbf{c}^{l,r}_{opt} = (\mathbf{V}\mathbf{R_s}\mathbf{V}^H + \mathbf{R_n})^{-1}\mathbf{V}\mathbf{R_s}[\mathbf{h}^{l,r}]^* \tag{7}$$

where $\mathbf{R_s} = E[\mathbf{s}\mathbf{s}^H]$, $\mathbf{R_n} = E[\mathbf{n}\mathbf{n}^H]$. Next, if $\mathbf{R_s}$ is unknown, one can assume that the sound field is diffuse as in [32], leading to $\mathbf{R_s} = \sigma_s^2 \mathbf{I}_L$ and with sources DOA uniformly spread around the sphere. Additionally, it is assumed that the noise is uncorrelated across microphones and identically distributed, such that $\mathbf{R_n} = \sigma_n^2 \mathbf{I}_M$, where σ_n^2 is the noise variance, σ_s^2 is the source variance and $\mathbf{I}_M$,$\mathbf{I}_L$ are the $M \times M$ and $L \times L$ identity matrices, respectively. These assumptions lead to the following simplification of Eq. (7) [16]:

$$\mathbf{c}^{l,r}_{BSM} = (\mathbf{V}\mathbf{V}^H + \frac{1}{SNR}\mathbf{I}_M)^{-1}\mathbf{V}[\mathbf{h}^{l,r}]^* \tag{8}$$

where $SNR = \frac{\sigma_s^2}{\sigma_n^2}$.

To reduce the negative impact of spatial cues at high frequencies becoming less accurate because of the loss of precision in BSM and increase in binaural error, it was suggested in [19] to modify BSM at high frequencies by minimizing the magnitudes in Eq. (5) using MagLS [18]. It was based on the idea that ILD is more noticeable than ITD for spatial perception at high frequencies [33]- [34]. In [32] it was shown that even with the use of MagLS the BSM still has a high error for high frequencies and that the performance rapidly degrades the further the microphones are from the ears of a virtual listener. In addition, in cases where the sound-field is directional, such as a speaker in a room, the diffuse sound-field assumption made in the development of the BSM fails which may lead to even higher errors. In this paper, methods that aim to improve the BSM method, especially in cases of directional sound-fields will be presented and analysed.

III. BSM Approaches that Incorporate Signal Information

This section introduces two distinct BSM-based binaural reproduction approaches. These approaches incorporate signal information, specifically modeling dominant sound sources, therefore approximating the solution of Eq. (7) more precisely, compared to the simplification of Eq. (8) which assumes a diffuse field. Throughout this section it is assumed that the sound-field, described in Eq. (1), is composed of direct and reverberant components. The direct component is composed of the direct sound from D far-field sources corresponding to a DOA of $\mathbf{\Omega_d} = [\Omega_1, ..., \Omega_D]^T$. The reverberant component is approximated to be composed of L far-field sources whose DOA's are uniformly spread on a sphere as in the model presented in Section II-B. This leads to the following formulation of the signal model:

$$\mathbf{x} = \mathbf{V_d}\mathbf{s_d} + \mathbf{V_r}\mathbf{s_r} + \mathbf{n} \tag{9}$$

where $\mathbf{V_d}$ is the $M \times D$ steering matrix of the direct component, $\mathbf{V_r}$ is the $M \times L$ assumed steering matrix for the reverberant component, and $\mathbf{s_d}$ and $\mathbf{s_r}$ are the source signal vectors for the direct and reverberant components, respectively. It is further assumed that the measurement noise $\mathbf{n}$ is white, uncorrelated to the source signals and that $\mathbb{E}[\mathbf{n}\mathbf{n}^H] = \sigma_n^2 \mathbf{I}_M$. Throughout this section, the following parameters are assumed to be separately estimated:

- The DOA of the D direct sources, $\mathbf{\Omega_d}$. Throughout this section, the dependency on $\mathbf{\Omega_d}$ for the direct component is omitted for simplicity.
- The measurements correlation matrix $\mathbf{R_x} = \mathbb{E}[\mathbf{x}\mathbf{x}^H]$.

It is also assumed that the source signals of the direct component are estimated by applying a linearly-constrained minimum-power (LCMP) beamformer [30], [35]:

$$\hat{\mathbf{s}}_\mathbf{d} = \mathbf{W_d}\mathbf{x} \tag{10}$$

where $\mathbf{W_d}$ is the weights matrix calculated using the constraint $\mathbf{W_d}\mathbf{V_d} = \mathbf{I}_D$ where $\mathbf{I}_D$ is a $D \times D$ identity matrix, as shown in [35] which leads to:

$$\mathbf{W_d} = (\mathbf{V_d}^H(\mathbf{R_x})^{-1}\mathbf{V_d})^{-1}\mathbf{V_d}^H(\mathbf{R_x})^{-1} \tag{11}$$

Note that because the LCMP is computed with correlation matrix $\mathbf{R_x}$, it is assumed that $\mathbf{s_d}$ is uncorrelated to $\mathbf{s_r}$ which is required to avoid signal cancellation [36].

A. BSM Incorporated with Parametric Spatial Coding

This subsection outlines a mathematical model that integrates a parametric spatial coding method with the BSM technique for an arbitrary array. This approach will be referred to as COMPASS-BSM (COM) which was previously formulated in [26], [27]. In this approach it is assumed that the measured signal can be described by Eq. (9). Using the estimated $\hat{\mathbf{s}}_d$, the binaural signal for the direct component can be calculated as follows Eq. (2):

$$\hat{p}_d^{l,r} = [\mathbf{h}_d^{l,r}]^T \hat{\mathbf{s}}_d \tag{12}$$

where $\hat{p}_d^{l,r}$ represents the binaural signal for the direct component at the ears and $\mathbf{h}_d^{l,r}$ denote the HRTF vectors associated with the direct component.

Previous studies have presented different methods for the rendering of the reverberant component [26], [37], [38]. The most common approach estimated the reverberant component directly by substituting Eq. (10) into Eq. (9), as:

$$\hat{\mathbf{x}}_\mathbf{r} = (\mathbf{I_M} - \mathbf{V_d}\mathbf{W_d})\mathbf{x} \tag{13}$$

Next, using $\hat{\mathbf{x}}_\mathbf{r}$, the reverberant component of the binaural signal, $\hat{p}_r^{l,r}$, can be estimated using the BSM method as in Eq. (3):

$$\hat{p}_r^{l,r} = (\mathbf{c}_{BSM}^{l,r})^H \hat{\mathbf{x}}_\mathbf{r} \tag{14}$$

where $\mathbf{c}_{BSM}^{l,r}$ are the standard BSM weights which are calculate as shown in Section II-B.

Finally, the binaural signal for the entire sound field is calculated as the sum of the binaural signals for the direct and reverberant components:

$$\hat{p}^{l,r} = \hat{p}_d^{l,r} + \hat{p}_r^{l,r} \tag{15}$$

where the resulting filter weights can be written by substituting Eqs. (12) and (14) into (15), as:

$$\mathbf{c}_{COM}^{l,r}(\mathbf{\Omega_d}, f) = (\mathbf{I_M} - \mathbf{V_d}\mathbf{W_d})^H \mathbf{c}_{BSM}^{l,r} + \mathbf{W_d}^H [\mathbf{h}_d^{l,r}]^* \tag{16}$$

where $\mathbf{c}_{BSM}^{l,r}$ are defined in Eq. (14), such that:

$$\hat{p}^{l,r} = (\mathbf{c}_{COM}^{l,r})^H \mathbf{x} \tag{17}$$

B. BSM with Informed Correlation Matrix

In this subsection, a BSM-based binaural reproduction approach that incorporates the direct component information into the source correlation matrix is introduced [20], [28] with the goal of better estimating the optimal filter described in Eq. (7). This approach will be referred to as Directional BSM (d-BSM). In this approach it

is assumed that the measured signal can be described by Eq. (9); that $\mathbf{s_d}$ and $\mathbf{s_r}$ are uncorrelated; and that $\mathbb{E}[\mathbf{s_r}\mathbf{s_r}^H] = \sigma_r^2\mathbf{I}_L$, where $\mathbf{I}_L$ is the $L \times L$ identity matrix and σ_r^2 is the assumed variance of the sources belonging to the reverberant component. Note that the latter assumption holds when the reverberant component is diffuse. Using $\hat{\mathbf{s}}_d$ as in Eq. (10) the covariance matrix of the sources belonging to the direct component can be estimated as:

$$\hat{\mathbf{R}}_{\mathbf{s_d}} = \mathbb{E}[\hat{\mathbf{s}}_d\hat{\mathbf{s}}_d^H] \tag{18}$$

Using $\hat{\mathbf{x}}_\mathbf{r}$ as in Eq. (13), the equivalent variance of the sources belonging to the reverberant component can be estimated as:

$$\hat{\sigma_r}^2 = \frac{1}{L}\|\hat{\mathbf{x}}_\mathbf{r}\|_2^2 \tag{19}$$

Next, the sources covariance matrix, $\mathbf{R_s}$, of dimensions $(D+L)\times(D+L)$ can be estimated from Eqs. (18) and (19) as:

$$\hat{\mathbf{R}}_\mathbf{s} = \begin{bmatrix} \hat{\mathbf{R}}_{\mathbf{s_d}} & \mathbf{0} \\ \mathbf{0} & \hat{\sigma_r}^2\mathbf{I}_L \end{bmatrix} \tag{20}$$

Formally, the variance of the sources should be added to that of the diffuse field originating from the source directions. However, this contribution is assumed negligible here, since the diffuse field energy is distributed over all directions, and thus becomes significant only at very low DRR. Now, by substituting $\hat{\mathbf{R}}_\mathbf{s}$ in Eq. (7), the optimal filter can be computed as:

$$\mathbf{c}_{d-BSM}^{l,r}(\mathbf{\Omega_d}, f) = \mathbf{B}^{-1}(\mathbf{V_d}\hat{\mathbf{R}}_{\mathbf{s_d}}[\mathbf{h}_d^{l,r}]^* + \hat{\sigma_r}^2\mathbf{V_r}[\mathbf{h}_r^{l,r}]^*) \tag{21}$$

where $\mathbf{h}_r^{l,r}$,$\mathbf{h}_d^{l,r}$ denote the HRTF vectors associated with the reverberant and direct components, respectively, $\mathbf{c}_{d-BSM}^{l,r}(\mathbf{\Omega_d})$ is the filter weight vector and $\mathbf{B}$ is calculated as:

$$\mathbf{B} = \hat{\sigma_r}^2\mathbf{V_r}\mathbf{V_r}^H + \mathbf{V_d}\hat{\mathbf{R}}_{\mathbf{s_d}}\mathbf{V_d}^H + \sigma_n^2\mathbf{I}_M \tag{22}$$

As in the standard BSM, at high frequencies, MagLS [18] can be used to minimize Eq. (5). Observing Eq. (5) it can be noted that it can equivalently be written as:

$$\begin{aligned} err_{bin}^{l,r}(\mathbf{\Omega_d}, f) = &\|\mathbf{R_s}^{0.5}([\mathbf{V}]^H\mathbf{c}^{l,r} - [\mathbf{h}^{l,r}]^*)\|_2^2 \\ &+ \|\mathbf{R_n}^{0.5}\mathbf{c}^{l,r}\|_2^2 \end{aligned} \tag{23}$$

Where $\mathbf{V} = [\mathbf{V_d}, \mathbf{V_r}]$ and $\mathbf{h}^{l,r} = [\mathbf{h_d}^{l,r}; \mathbf{h_r}^{l,r}]$. By substituting Eq. (18) into Eq. (23) the MagLS formulation can be written as:

$$\begin{aligned} \mathbf{c}_{d\text{-}BSM}^{l,r}(\mathbf{\Omega_d}, f) = \arg\min_{\mathbf{c}} &\left\| |\hat{\mathbf{R}}_\mathbf{s}^{0.5}\mathbf{V}^H\mathbf{c}^{l,r}| - |\hat{\mathbf{R}}_\mathbf{s}^{0.5}[\mathbf{h}^{l,r}]^*| \right\|_2^2 \\ &+ \sigma_n^2\|\mathbf{c}^{l,r}\|^2 \end{aligned} \tag{24}$$

C. Distinction between the two approaches

This subsection aims to provide some insight into the differences between the two approaches. The COM method separates direct and reverberant components to provide a better model for the reverberant component but may be sensitive to estimation errors. Conversely, the d-BSM method estimates only the variance of the reverberant component without needing to estimate this component's signal. It is therefore expected to be more robust against estimation errors. Essentially, the trade-off between the two methods lies in the detail of modeling and robustness to estimation error.

IV. Error Measures

This section presents the primary error metrics employed throughout this study to assess and compare the various binaural reproduction techniques.

A. Normalized Mean-Square Error

The normalized mean-squared error (NMSE) of the binaural signals is calculated as:

$$NMSE^{l,r}(f) = \begin{cases} \frac{|\hat{p}^{l,r}(f) - p_{ref}^{l,r}(f)|^2}{|p_{ref}^{l,r}(f)|^2}, & \text{for } f < 1.5\text{kHz} \\ \frac{||\hat{p}^{l,r}(f)| - |p_{ref}^{l,r}(f)||^2}{|p_{ref}^{l,r}(f)|^2}, & \text{for } f \geq 1.5\text{kHz} \end{cases} \tag{25}$$

where $p_{ref}^{l,r}(f)$ is the reference binaural signal of a certain frequency, and $\hat{p}^{l,r}(f)$ is the reproduced binaural signal of the same frequency. For frequencies above 1.5 kHz the NMSE of the absolute value was calculated based on magnitude only, in accordance with the duplex theory, which states that localization at high frequencies is governed primarily by magnitude-based cues, as shown in [39].

B. Log Spectral Distance

The Log Spectral Distance (LSD) is a common measure to represent averaged differences in magnitude [40], computed here for the binaural signals as:

$$LSD^{l,r}(t) = \sqrt{\frac{1}{N_{ERB}}\sum_{n=0}^{N_{ERB}}\left(10\log_{10}\frac{|p_{\text{test}}^{l,r}(f,t)|^2|G_n(f,f_c)|}{|p_{\text{ref}}^{l,r}(f,t)|^2|G_n(f,f_c)|}\right)} \tag{26}$$

where $p_{ref}^{l,r}$ and $p_{test}^{l,r}$ represent the binaural reference and the signal under test in the time-frequency domain, and $G_n(f, f_c)$ represents the gammatone filter of the Equivalent Rectangular Bandwidth (ERB) filter bank, centered at frequency f_c. An average of 14 filter bands in the $[1.5, 8]$ kHz range, generated using the Auditory Toolbox [41], was used to compute the frame-based $LSD^{l,r}(t)$ error. The total LSD error is obtained by

averaging N_T frames of the LSD frame-based error over all signals frames:

$$LSD^{l,r} = \frac{1}{N_T}\sum_{t=1}^{N_T} LSD^{l,r}(t) \tag{27}$$

C. ITD and ILD

In this subsection, performance measures based on ITD and ILD are outlined. Throughout this section, it is assumed that the sound-field, as described in Section II-A, comprises solely of a single source, having a DOA denoted as Ω, with a unit variance. Using Eq. (3), both $p^l(t)$ and $p^r(t)$, representing the left and right time-domain binaural signals, were computed for a chosen BSM filter. The reference binaural signals were derived by substituting the source signal in Eq. (2).

The ITD computation employed a well-accepted perceptual measure [42], utilizing a cross-correlation based method. Initially, the binaural signals were low-pass filtered at a cut-off frequency of $1.5\,\text{kHz}$, followed by the ITD calculation depicted as:

$$ITD(\Omega) = \arg\max_{\tau} |\sum_{t=0}^{T-\tau-1} p^r(t)p^l(t+\tau)| \tag{28}$$

where T represents the total number of time samples. The chosen cutoff frequency of $1.5\,\text{kHz}$ aligns with the frequency range crucial for spatial perception through phase or binaural time differences.

After computing ITD for both the BSM filter and the reference signal, the ITD error measure was computed as:

$$\epsilon_{ITD}(\Omega) = |ITD(\Omega) - ITD_{ref}(\Omega)| \tag{29}$$

For the ILD estimation, the binaural signals was processed using ERB filter bands, as outlined in [43]:

$$ILD_f(f_c,\Omega) = 10\log_{10}\frac{\sum_{f=0}^{f_c^{max}} |G(f,f_c)p^l(f)|^2}{\sum_{f=0}^{f_c^{max}} |G(f,f_c)p^r(f)|^2} \tag{30}$$

where $G(f,f_c)$ represents the ERB filter with central frequency f_c at frequency f, and f_c^{max} denotes the maximum frequency of this ERB filter. An average of 14 filter bands in the $[1.5, 8]\,\text{kHz}$ range were used to compute ILD as:

$$ILD(\Omega) = \frac{1}{14}\sum_{f_c} ILD_f(f_c,\Omega) \tag{31}$$

Subsequently, the ILD error measure was computed as:

$$\epsilon_{ILD}(\Omega) = \frac{1}{14}\sum_{f_c} |ILD_f(f_c,\Omega) - ILD_{ref}(f_c,\Omega)| \tag{32}$$

V. Simulation Study - Setup and Methodology

In this section, a comparison between the two approaches, presented in Section III, and the standard BSM, presented in Section II-B, is provided. Broadly, this section aims to study the benefits and to gain insights into signal-dependent BSM approaches under various conditions, including robustness to estimation error.

A. Simulation Set-Up

In this section, we employ a four-microphone array mounted on a rigid sphere. The microphone layout was inspired by the AR-glasses geometry provided in the EasyCom dataset [29], [44]. The simulated array excluded the two ear microphones of the EasyCom array, while microphone positions were projected onto a constant-radius sphere to simplify the simulation setup. The microphone positions were denoted using spherical coordinates (r, θ_m, ϕ_m) for $m = 1, \ldots, M$ relative to the array center, as detailed in Table I

Parameter	Value
M (microphones)	4
r (m)	0.08
θ_m (deg)	[82, 81, 82, 80]
ϕ_m (deg)	[53,0,320,290]

TABLE I
ARRAY CONFIGURATION WITH MICROPHONE POSITIONS ON A RIGID SPHERE, INSPIRED BY THE AR-GLASSES GEOMETRY REPORTED IN THE EASYCOM DATASET [29].

The steering vectors of this array were computed analytically in the SH domain following section 4.2 in [45] using the array geometry and the DOAs of the far-field sources modeled as plane waves. The HRTF in the simulation was obtained from the Cologne database measurements of the Neumann KU100 manikin [46]. The HRTFs for the DOAs of the assumed sources were interpolated in the SH domain using a SH order of 30. The assumed listener head was aligned with the positive x axis. An illustration of the array position relative to the head position and alignment is presented in Figure 1.

To evaluate performance across a broad range of acoustic scenarios, a database consisting of over 260 Monte Carlo simulations was constructed. In each simulation, a point source was located within a room. The microphone array was placed inside the room, facing the x-axis, as depicted in Figure 1(a). Other simulation parameters were randomly selected from the values specified in Table II. In particular, listener's head rotations,

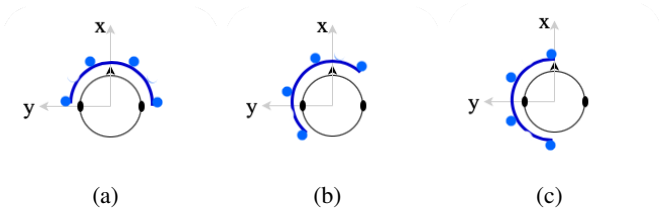


Fig. 1. An illustration of a virtual listener head with a 4-microphone array assuming: (a) 0° array rotation, (b) 50° array rotation, and (c) 90° array rotation relative to the listener's head. The blue dots represent the microphones , and the x and y gray arrows represent the positive x and y axes, respectively.

imposed during binaural reproduction (not during recoding) are illustrated in Figs. 1(b), 1(c) with a relative angle between the array and the listener's head. The transfer function from the source to each microphone in the array was simulated using the image source method [47], with impulse responses truncated to the duration of the room reverberation time, T_{60}.

Parameter	**Value**
Source DOA (deg)	$[0, 15, 30, 60]$
Clean speech source	4 Male, 4 Female
Source signal duration (s)	$[2.5 - 10]$
Rooms dimensions (m)	$(5 \times 4 \times 3)$, $(10 \times 8 \times 3.5)$, $(20 \times 15 \times 6)$
Array position (m)	$(1.66, 2, 1.8)$, $(3.33, 4, 1.8)$, $(7.91, 9.66, 1.8)$
Source distance from array (m)	$(0.13 - 6)$
T_{60} (s)	$[0.3, 0.6, 1.4]$
DOA Error (deg)	$[0, 2, 4, 6, 10, 30]$
Head Rotation (deg)	$[0, 30, 60, 90]$

TABLE II
MONTE CARLO ROOM SIMULATION PARAMETERS

Clean speech recordings taken from the TIMIT and WSJ0 databases [48], [49]. All signals were sampled at 16 kHz.

B. Methodology

In this study, the BSM filter $\mathbf{c}_{BSM}^{l,r}$ was computed following Eq. 8 as detailed in Section II-B, and MagLS [18] was employed for frequencies within the range of $[1.5, 8]$ kHz. The steering matrix was computed assuming $L = 400$ sources nearly-uniformly distributed on a sphere according to [50], as elaborated in Section II-B. The MagLS solution was obtained using the method suggested in [27]. The steering vector of the direct component with a DOA of Ω_d was computed from the source and array positions (see Table II).

Subsequently, after generating microphone signals, $\mathbf{x}(t)$, as described in section V-A, $\mathbf{x}[t, f]$ was computed using Short-Time Fourier Transform (STFT) with a Bartlett window of duration 0.032 s and a hop size of 0.008 s. Following this, the measurements' correlation matrix $\mathbf{R_x}$ was computed by averaging across both time and frequency domains according to:

$$\hat{\mathbf{R}}_{\mathbf{x}}[f] = \frac{1}{T} \sum_{t=1}^{T} \sum_{j=-J}^{J} \mathbf{x}[t, f + j \cdot \Delta f] \mathbf{x}^H[t, f + j \cdot \Delta f] \tag{33}$$

where T denotes the total length in seconds of the recording, Δf is the frequency resolution and J controls the bandwidth of the frequency smoothing. Subsequently, the LCMP beamformer was computed following Eq. (11), leading to the determination of $\mathbf{c}_{COM}^{l,r}(\mathbf{\Omega}_d, f)$ using $\mathbf{c}_{BSM}^{l,r}$, as per Eq. (16) in Section III-A. Further, $\mathbf{c}_{d-BSM}^{l,r}(\mathbf{\Omega}_d, f)$ was calculated following the equations outlined in Section III-B, employing the BSM steering matrix as the reverberant component's steering matrix and utilizing MagLS [18] for frequencies within the range of $[1.5, 8]$,kHz. Following the computation of the three filters, the binaural signals for all filters were computed using Eq. (3) by replacing the appropriate filter $\mathbf{c}^{l,r}$. The reference binaural signals were obtained by convolving the HRTFs of the left and right ears with the HOA signals of order 14, which were computed using the image method as described in Section V-A. Finally, the errors measures as presented in Section IV were computed using the binaural signals for the three methods, and the reference binaural signals.

VI. SIMULATION STUDY - RESULTS

In this section, the entire database will first be evaluated under nominal conditions, i.e. without DOA estimation errors and head rotation. Subsequently, these will be incorporated to assess performance in more challenging scenarios. The values corresponding to these conditions are detailed in the last two rows of Table II. In addition, performance is evaluated across varying direct-to-reverberant ratio (DRR) conditions, and through the analysis of ITD and ILD. It is important to note that evaluations in this study are conducted under single-source conditions. This choice, which can be considered as a simplification compared with prior multi-source evaluations of similar parametric methods (e.g., [24], [26], [27]), enables an unambiguous and isolated assessment of the methods' fundamental performance limits across varying acoustic conditions.

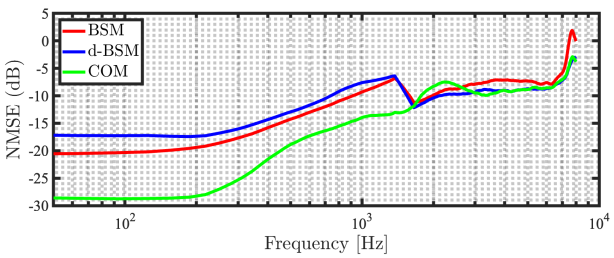


Fig. 2. The overall NMSE of the binaural signal, averaged across both ears and over all Monte Carlo simulations under nominal conditions, presented for the three methods. The NMSE values are computed as defined in Eq. (25), i.e. with a combined least-squares and magnitude least-squares along frequency, assuming a signal-to-noise ratio (SNR) of 20dB, and normalized at each frequency to ensure minimal dependence on the source signal.

A. Analysis under Nominal Conditions

Performance of the various methods was first evaluated using the metrics described in Section IV under nominal conditions, i.e. with no head rotation and no DOA error. NMSE and LSD values were averaged across all scenarios in the database, and are presented in Table III and Fig. 2.

Method	$LSD^{(l)}$	$LSD^{(r)}$	$NMSE^{(l)}$	$NMSE^{(r)}$
BSM	1.75	1.45	-8.4	-7.2
d-BSM	2.09	1.83	-9.6	-8.2
COM	**0.96**	**0.87**	**-10.6**	**-9.4**

TABLE III
NMSE AND LSD VALUES AVERAGED OVER ALL MONTE CARLO SIMULATIONS. ALL ERRORS ARE MEASURED IN [dB].

Table III shows that the COM and d-BSM methods outperforms BSM, with COM showing the best performance, but the differences are not large when averaged across all conditions in the database. Furthermore, Fig. 2 shows that up to about 1.5 kHz, the COM method demonstrates a notable advantage over the other approaches, while at higher frequencies, d-BSM seems to outperform the other two methods. These COM-BSM performance trends align with earlier findings [26], which similarly observed that COM consistently surpasses BSM across a range of acoustic conditions. Furthermore, as previously suggested, the EasyCom geometry may pose challenges for parametric rendering methods [26], reflected here by the elevated d-BSM LSD values. Nevertheless, the overall results indicate that parametric approaches remain robust and effective even under such suboptimal array configurations.

B. Performance as a function of DRR

In this subsection, the NMSE and LSD measures are evaluated as a function of DRR. The analysis is conducted over the Monte Carlo database, under nominal conditions. For this analysis, four DRR value ranges were first defined, and each acoustic scene was classified into one of these ranges according to the DRR value of the scene.

	LSD (dB)			**NMSE (dB)**		
DRR Range (dB)	**BSM**	**d-BSM**	**COM**	**BSM**	**d-BSM**	**COM**
[-11.0 -4.5)	1.54	2.63	**1.29**	-6.4	-5.6	**-6.5**
[-4.5 2.5)	1.50	2.45	**0.99**	-7.0	-6.6	**-8.0**
[2.5 9.0)	1.60	1.46	**0.70**	-8.4	-10.3	**-11.4**
[9.0 15.0]	1.79	1.29	**0.75**	-9.5	-13.4	**-14.3**

TABLE IV
Averaged LSD and NMSE values in decibels, divided into DRR ranges.

Table IV presents the averaged NMSE and LSD results across the different DRR classes. At low DRR values [-11 to -4.5] dB, where reverberant signal components are more dominant, the performance of d-BSM and COM is similar to BSM. This can be explained by conditions that are similar to the design assumption of a diffuse field [16]. In contrast, at higher DRR values [9 to 15] dB, d-BSM and COM significantly outperform BSM, by up to 5dB. Here as well COM presents the best results throughout the various conditions. These findings are consistent with earlier observations [26], which likewise identified COM as the best-performing parametric method across diverse acoustic scenarios, and thereby underscore the benefit of incorporating signal-dependent information into binaural reproduction under more directional, source-dominant conditions.

C. Robustness to DOA Estimation Errors

In this subsection, performance is evaluated assuming an error in the estimation of source direction, required for the d-BSM and COM methods. Maintaining performance under such conditions is important as some estimation error is expected in practice. Assuming Ω_d is the true DOA of the source, the methods under study will be employed with an estimated source DOA of $\Omega_d + \Omega_{err}$ where Ω_{err} is the error in the DOA estimation.

Table V displays the NMSE and LSD metrics for all methods. The table shows that d-BSM and COM methods, being dependent on accurate estimation of the direct source DOA, are particularly sensitive to such errors.

In particular, relatively small deviations in DOA of a few degrees can lead to substantial performance degrada-

tion, particularly for the COM method. Under large DOA errors, the performance of the d-BSM and COM methods tends to converge with the BSM method. Although these results demonstrate the sensitivity of d-BSM and COM to DOA estimation errors, their performance does not degrade much below that of BSM.

	LSD (dB)			NMSE (dB)		
DOA Err [°]	**BSM**	**d-BSM**	**COM**	**BSM**	**d-BSM**	**COM**
0	1.60	1.97	**0.92**	-7.8	-8.9	**-10.0**
2	1.60	2.03	**1.51**	-7.8	**-7.9**	-7.4
4	1.60	2.05	**1.55**	**-7.8**	-7.7	-7.4
6	1.60	2.05	**1.55**	**-7.8**	-7.5	-7.4
10	1.60	2.05	**1.56**	**-7.8**	-7.4	-7.5
30	**1.60**	2.03	**1.60**	**-7.8**	-7.4	-7.6

TABLE V
Averaged LSD and NMSE values in decibels as function of DOA Error in degrees.

D. Robustness to Head Rotation

In this subsection, the robustness of the NMSE and LSD metrics to listener head rotations is evaluated, for head rotations of 30, 60 and 90 degrees. The metrics are presented as average over the two ears, and detailed in Table VI. The table shows that both d-BSM and COM methods exhibit greater robustness to head rotation compared to BSM. As the rotation angle increases, the performance degrades, even with MagLS [32], [17], whereas d-BSM shows moderate robustness. The COM method, in particular, maintains consistently low LSD error, indicating strong resilience to magnitude errors. Overall, the results of the d-BSM and COM are better suited for listening conditions with listener head movements, which also reinforce the dynamic head-rotation findings reported in [27].

	LSD (dB)			NMSE (dB)		
Head Rot [°]	**BSM**	**d-BSM**	**COM**	**BSM**	**d-BSM**	**COM**
0	1.60	1.97	**0.92**	-7.8	-8.9	**-10.0**
30	2.11	2.33	**1.14**	-6.6	-9.0	**-9.6**
60	2.73	2.64	**1.42**	-6.0	-9.2	**-9.4**
90	2.85	2.65	**1.54**	-6.0	-9.0	**-9.1**

TABLE VI
Average LSD and NMSE values in decibels as function of Head Rotation values in degrees.

E. Analysis of ITD and ILD

Performance metrics studied thus far are based on NMSE and LSD. However, this might not fully capture human spatial perception. Therefore, in this sub-section the study employs perceptually driven measures, ITD and ILD, as in Sec. IV-C. As an illustrative example,

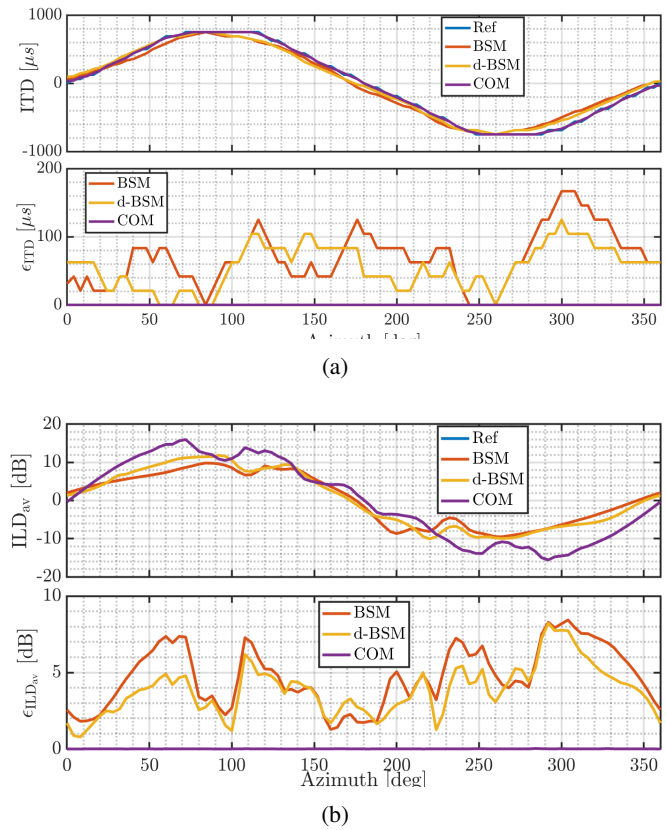


Fig. 3. The ITD (a) and ILD (b) were evaluated using the direct sound component of the scenario selected in IV-C. The red line represents the BSM method, the yellow line corresponds to the d-BSM approach, and the purple line depicts the COM approach. ITD and ILD errors are also presented, as defined in Section IV-C. No head rotation was employed.

a single scenario was selected from the following parameters: room dimensions of [5, 4, 3] (m), $T_{60} = 0.3$ (s), DRR: 14.8 (dB), array position of [1.66, 2, 1.8] (m), and a female speech source signal of 4 (s). The calculation of the ILD and ITD has been divided into two parts. First, a source in a room was simulated generating according to the selected scenario. The recorded signals at the array were then used to compute filters for the methods as described in Section V-B. Now, using the same filters, binaural signals were computed but this time for a sound field composed of the same source in free-field, with the same direction of arrival as the source in the room as in the selected scenario. Then, ILD and ITD were calculated as defined in Section IV-C, for this specific source direction. Finally, the same process of computing filters and then ILD and ITD, was repeated for a range of source directions to complete the ILD and ITD analysis. This somewhat non-standard analysis was employed so that the filters could be computed using

realistic signals, but then evaluated in terms of ILD and ITD in the direction of the source only, as the direct sound is known to dominate sound localization.

Figure 3 illustrates the performance of ILD and ITD as detailed above for the three methods. The source direction was set on the horizontal plane with $\theta = 90°$, and no rotation of the head was considered. The figure shows that all three methods perform reasonably well in terms of ITD, and for most directions remain below the just-notable difference (JND) of $100\mu s$ [51]. Furthermore, both d-BSM and COM outperformed BSM in terms of ILD, with the COM method demonstrating the highest performance, maintaining errors well below the JND threshold of 1dB [51]. Similar trends were reported in [26], which align with the present findings and further validate the strong performance of COM, particularly under high DRR conditions. This outcome particularly highlights the strong potential to leverage signal information in binaural signal matching.

VII. Listening Experiment

The simulation results in the previous section offer valuable insights into the objective performance of the different BSM approaches, but they do not fully capture how these differences are perceived by human listeners. This section describes a listening experiment designed to subjectively compare the quality of the BSM, d-BSM, and COM methods.

A. Setup

The experimental setup was based on a simulation of a point source inside a room. The source was positioned at an angle of $\Omega_d = (90°, 30°)$ with the array front-looking direction at $(90°, 0°)$, which was also simulated as described in Section V-A, i.e., an array of 4 microphones mounted on a rigid sphere in a similar configuration to the wearable EasyCom array. The room impulse responses from the source to the array microphones were simulated using the image method [47] in MATLAB [52] rather than recorded impulse responses. While recorded responses offer acoustic authenticity, the image method was selected to maintain precise, independent control over specific acoustic parameters, allowing us to strictly regulate the reverberation time , DRR, and source angles without the confounding variables often introduced by measurement inconsistencies in real-world environments.

Microphone signals were generated by convolving the simulated impulse responses with speech signals selected from Table II. The binaural synthesis was performed using non-individualized HRTFs. Specifically, the measured HRTFs from the Cologne HRTF database corresponding to the Neumann KU100 dummy head were utilized [46]. Measuring individual HRTFs for each participant was not feasible within the scope of this study, however, the KU100 dataset provides a widely accepted reference for generic binaural reproduction.

The study employed the Multiple Stimuli with Hidden Reference and Anchor (MUSHRA) test [53]. Four independent variables (within-subject factors) were manipulated:

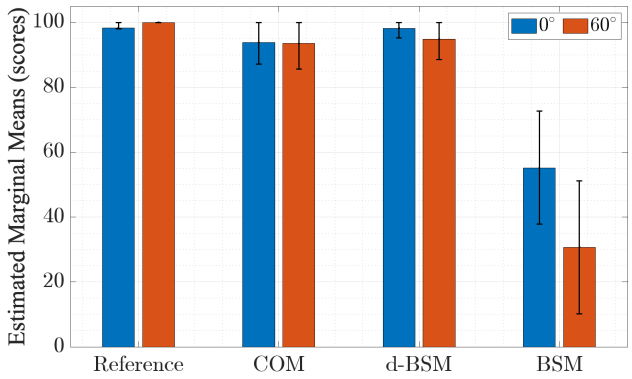
- **Stimuli**: male and female speech.
- **Head rotation angles**: $0°$ and $60°$.
- **Rooms**: a small room ($[5, 4, 3]$ m, $T_{60} = 0.3$ s) and a medium room ($[10, 8, 3.5]$ m, $T_{60} = 0.6$ s).
- **Methods**: Ref, COM, d-BSM, BSM.

The array was placed at $[1.6, 2, 1.8]$ m in the small room and at $[3.3, 4, 1.8]$ m in the medium room. Source positions were chosen such that the DRR was $+14$ dB in the small room and 0 dB in the medium room. The small room thus simulated a low-reverberation environment with dominant direct sound, while the medium room provided a more diffuse sound field with stronger early and late reflections. Combining the three factors with two levels each resulted in 8 MUSHRA screens.

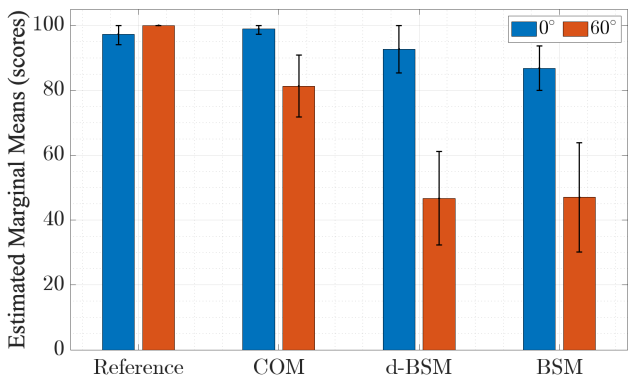
B. Methodology

Binaural signals were generated as described in Section V-B. Each MUSHRA screen included four test signals: the hidden reference (HOA with $N = 14$), BSM, d-BSM, and COM, implemented as described in Section V-B. It should be noted that while standard MUSHRA protocols [53] recommend the use of a low-pass filtered anchor, an explicit anchor was not included in this experiment. This exclusion was intentional to avoid the potential bias of an inappropriate anchor widening the ranking range, which could unnecessarily compress the subjective differences between the specific test signals under investigation.

Nine subjects, all reporting normal hearing, participated in the study. The MUSHRA screens and signals were presented in randomized order. Participants rated the similarity of each test signal to the reference on overall quality, including both spatial and timbral quality. Scores ranged from 0 to 100, with 100 indicating that the test signal was indistinguishable from the reference. Headphone compensation filters cited in [46] were applied. Prior to the listening test, participants underwent a training stage to familiarize themselves with the scoring procedure.



(a) Small Room with DRR=+14dB



(b) Medium Room with DRR=0dB

Fig. 4. Listening test results. The estimated marginal means with 95% confidence intervals are shown for the COM, d-BSM, and BSM methods. Overall quality ratings are presented for both the non-rotated (blue, 0°) and rotated (orange, 60°) conditions. The top figures correspond to the small room, while the bottom figures show results for the medium room; in both cases, ratings are averaged across stimuli.

C. Results

A repeated-measures analysis of variance (RM-ANOVA) [54] with four within-subject factors and their interactions was performed on the collected scores. Mauchly's test indicated minor violations of sphericity, with $\epsilon > 0.75$ for most factors and interactions, and a moderate violation ($\epsilon > 0.4$) for the method–angle–room and room–method interactions; accordingly, the Greenhouse-Geisser correction was applied. The analysis revealed statistically significant main effects for head rotation angle, $F(1,9) = 35.8, p = .002, \eta_p^2 = .88$, method, $F(3,27) = 22.9, p = .002, \eta_p^2 = .82$, and room, $F(1,9) = 5.56, p = .046, \eta_p^2 = .41$. Furthermore, the three-way interaction between room, head rotation angle, and method was significant, $F(3,27) = 6.2, p = .018, \eta_p^2 = .44$.

To further examine the effects of room and head rotation, post-hoc pairwise comparisons with Bonferroni correction were conducted between the estimated marginal means of the reference and the evaluated methods. The estimated marginal mean scores with 95% confidence intervals from the participant ratings are presented in Fig. 4. The results can be summarized as follows:

- **Small room, no rotation:** The mean differences between the reference and the other methods were 5.57, 1.18, and 44.22 points for COM, d-BSM, and BSM, respectively, with $p = 0.68$, 1.0, and 0.003. These results suggest that when the head is stationary and the direct signal dominates over reflections (high DRR), parametric methods produce binaural signals highly similar to HOA, while the signal-independent BSM results is signals that are noticeably different from the HOA reference.
- **Medium room, no rotation:** The mean differences were 1.55, 4.62, and 10.50 points for COM, d-BSM, and BSM, respectively, with $p = 1.0$, 1.0, and 0.003. In scenarios with lower DRR, where reflections are more pronounced, the performance of BSM and parametric methods is reduced, but all methods still produce signals resembling the HOA reference in terms of overall quality.
- **Small room, 60° rotation:** The mean differences were 6.50, 5.12, and 69.22 points for COM, d-BSM, and BSM, respectively, with $p = 0.55$, 0.59, and 0.01. These results indicate that in high DRR conditions, the accuracy of parametric methods remains largely unaffected, while BSM shows noticeable degradation.
- **Medium room, 60° rotation:** The mean differences were 18.61, 53.22, and 52.94 points for COM, d-BSM, and BSM, respectively, with $p = 0.01$, 0.01, and < 0.01. These findings highlight the accuracy degradation induced by head rotation in low DRR scenarios. BSM and d-BSM exhibit the largest reduction in overall quality, whereas the COM method demonstrates the greatest robustness to head rotations.

Overall, the results validate the simulation study results and demonstrate that head rotation and room acoustics both significantly impact perceived binaural quality. The BSM method consistently exhibited the greatest degradation in overall quality under head rotation, particularly in low-DRR scenarios, while d-BSM showed moderate reductions. In contrast, the COM method

proved the most robust across all conditions, maintaining high similarity to the HOA reference even when the head was rotated or in rooms with more pronounced reflections. These findings highlight the resilience of COM to both head movement and room reverberation compared to the other methods evaluated. Given that COM and BSM were extensively analyzed in earlier perceptual studies [26], [27], they form the natural basis for comparison with the present work. Consistent with these findings, our results show that COM again yields small perceptual deviation from the reference in both high DRR and reverberant conditions [26]. This trend also holds under listener rotation, further reinforcing the conclusion that COM offers the most robust reproduction across orientations compared to signal-independent BSM [27]. The newly proposed d-BSM method, though absent from earlier studies, fits naturally within this comparison. Under high DRR conditions, d-BSM performs similarly to COM, demonstrating its strength when the direct sound dominates. Its performance decreases in more reverberant environments and under listener head rotation, which may be attributed to the rendering technique of the reverberant component in Eq. (21).

VIII. Conclusions

This paper offers several significant contributions to the field of binaural reproduction and its application using an array configuration motivated by that of a wearable array. The paper analyzes and compares two approaches that incorporate signal information into the BSM method, specifically focusing on source DOA. These methods effectively address the limitations imposed by the diffuse-field assumption of BSM, demonstrating improved performance under various conditions, and provide additional validation that reinforces findings from previous studies conducted under less diverse acoustic conditions.

In particular, improvement was shown to be significant at high DRR conditions, for which the direct sound from the source is dominant, and with listener head rotation which was shown to significantly degrade the performance of BSM. An error analysis revealed good robustness of both methods under study to errors in DOA estimation, which highlights the potential of these methods especially for d-BSM. in practical scenarios where such error is expected. The limitation of the work is that the methods were primarily tested in controlled environments, and so analysis of their performance in more complex real-world scenarios and in the presence of multiple sources is proposed for future work.

IX. Declarations

A. Availability of data and materials

The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

B. Competing interests

The authors declare that they have no competing interests.

C. Funding

This research was supported in part by Reality Labs Research at Meta and Ben-Gurion University of the Negev.

D. Authors' contributions

AB performed the entire research and wrote the original version of the paper. MY performed the simulation in the revised manuscript. OB performed the listening test in the revised manuscript. VT, JD, ZBH, and BR supervised the research. The authors read and approved the final manuscript.

E. Authors' information (optional)

Not applicable

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